

DP-CBS (depth-domain Duato)

pooled window $[r, V)$
may congest arbitrarily (cap S only)

↓ always eligible,
never capped

dedicated window $[0, r)$
private CBS sub-ring keeps its bubble

*transit may fill the last free dedicated slot
only from a dedicated VC (bubble never leaks to the pool)*

DP-ESC (Duato verbatim)

adaptive window $[0, V - 1)$
fully pooled, may congest arbitrarily

↓ always eligible,
never capped

escape VC $(V - 1)$
exempt from pool; acyclic Mesh3D XYZ

*escape CDG untouched by DP
⇒ Duato's theorem applies unchanged*

Safety composition: the pool is never in the deadlock-freedom path